

The problem: Localhost agents cannot be shared

Introduction

You've built sophisticated agents:

- Course 1-3: Basic agents with models, tools, instructions
- Course 4: State management with 4 namespaces
- Course 5: Advanced tool integration
- Course 6: Guardrails with callbacks
- Course 7: Multi-agent coordination
- Course 8: Complex workflows with A2A protocol (localhost)

But there's a critical limitation:

```
Python
# From Courses 1-8: Everything runs on localhost
# Terminal 1: adk web
# Access: http://localhost:8000

# Problem: Only accessible on YOUR computer
# Cannot share with team or users
# Stops when your computer sleeps
# No 24/7 availability
```

This course solves the “localhost gap” by teaching you cloud deployment fundamentals.



The problem:

Localhost agents cannot be shared

Consider a sophisticated agent:

Complex workflow:

Python

```
# Content QA system (works locally)
```

```
from google.adk.agents import SequentialAgent, ParallelAgent, LoopAgent
```

```
workflow = SequentialAgent(  
    sub_agents=[  
        intake,  
        ParallelAgent(sub_agents=[validator_a, validator_b, validator_c]),  
        LoopAgent(sub_agents=[quality_check, refiner], max_iterations=5)  
    ]  
)
```

```
# Testing: adk web
```

```
# Access: http://localhost:8000
```

```
# Problem: Only accessible on my computer
```


The fundamental limitations:

✗ Cannot share with team or users

- Localhost URL (<http://localhost:8000>) only works on your machine
- Team members cannot access your agent
- Users cannot test or use your agent
- No way to integrate into production systems

✗ No scalability

- Single instance on your computer
- Cannot handle multiple concurrent users
- Performance limited by local machine
- No automatic scaling

✗ No 24/7 availability

- Agent stops when you close terminal
- Agent stops when computer sleeps or restarts
- Cannot serve users in different time zones
- No reliability for production use

✗ No production persistence

- InMemorySessionService loses data on restart
- Session state not preserved across deployments
- No cross-session memory
- Cannot learn from past interactions

The root problem

Localhost development is perfect for building and testing, but production systems need cloud deployment for sharing, reliability, and scalability.

